LAB-III TASK 2 lab Report

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Section B

Title:

Use of Logical Operators in C

Objective:

To check whether a number lies within a specific range using logical operators.

Theory:

Logical operators are used to combine multiple conditions:

- `&&` (AND) → true if both conditions are true
- `||` (OR) → true if at least one condition is true
- `!` (NOT) → reverses the result

Program (with comments):

```
#include <stdio.h>   // Input-output library

int main()
{
    int num;  // Declare integer variable

    printf("Enter a number: "); // Ask user for input
    scanf("%d", &num);          // Store input

    // Check if number is between 0 and 100
    if (num > 0 && num < 100)   // Logical AND condition
    {
        printf("Number is between 0 and 100.\n");
    }
    else                        // If condition is false
    {
        printf("Number is NOT between 0 and 100.\n");
    }

    return 0; // End program
}
```

Explanation:

The program uses the logical AND (`&&`) operator to check if both conditions are true simultaneously.

Conclusion:

Logical operators are essential for combining multiple conditions in decision-making.